

## Proximal Point Algorithm Converges

$$\min: f(x)$$

$$x_{t+1} \leftarrow \text{prox}_{\lambda f}(x_t) = \arg\min_x f(x) + \frac{1}{2\lambda} \|x - x_t\|^2$$

$1/\lambda$  convergence rate

Recall: Prox Grad smooth

$$f(x) = g(x) + h(x)$$

$$\text{Here: } f(x) = 0 + f(x)$$

$$\text{Prox Grad: } x_{t+1} \leftarrow \text{prox}_{\lambda f}(x_t - \cancel{y_t} \nabla f(x_t))$$

## Proximal Point Algorithm Converges - continued -

$$\begin{aligned}
 x_{t+1} &= \text{prox}_{\lambda h} (x_t - \gamma \nabla g(x_t)) \\
 &= x_t - \lambda G_\lambda(x_t), \quad G_\lambda(x_t) = \frac{1}{\lambda} (x_t - \text{prox}_{\lambda h} (x_t - \gamma \nabla g(x_t)))
 \end{aligned}$$

For our setting:  $f(x) = h(x)$ ,  $g = 0$ .

$$G_\lambda(x) = \frac{1}{\lambda} (x_t - \text{prox}_{\lambda f} (x_t))$$

$$\begin{aligned}
 f(x_{t+1}) - f(x^*) &\leq \langle G_\lambda(x_t), x_t - x^* \rangle - \frac{\lambda}{2} \|G_\lambda(x_t)\|_2^2 \\
 &= \frac{1}{2\lambda} \left( \|x_t - x^*\|_2^2 - \underbrace{\|x_t - x^* - \lambda G_\lambda(x_t)\|_2^2}_{\lambda^2 \|G_\lambda(x_t)\|_2^2} \right)
 \end{aligned}$$

$$= \frac{1}{2\lambda} (\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2)$$

$$\Rightarrow \lambda_t (f(x_{t+1}) - f(x^*)) \leq \frac{1}{2} (\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2)$$

$$\text{Sum to conclude: } f(x_T) - f(x^*) \leq \frac{\|x_0 - x^*\|^2}{2 \sum \lambda_t}$$

## Accelerated PPA

Use "FISTA" like acceleration

Variation from accelerated ~~prox~~ prox grad

- Initialize  $x_0$

- $x_1 = \text{prox}_{\lambda_0 f}(x_0)$

- $x_{t+1} = \text{prox}_{\lambda_t f}(x_t + \theta_t(\frac{1}{\theta_t} - 1)(x_t - x_{t-1}))$

Choose  $\lambda_t$ , and then  $\theta_t$  that solves

$$\frac{\theta_t^2}{\lambda_t} = (1 - \theta_t) \frac{\theta_{t-1}^2}{\lambda_{t-1}}$$

$\rightarrow \sum \sqrt{\lambda_t} \rightarrow \infty$

For fixed  $\lambda_t = \lambda$  :  $O(1/\sqrt{T})$  convergence

## Fixed Point Mappings

$$\text{prox}_{\lambda f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

Fixed Point Property:

$\hat{z}$  a fixed point of  $\text{prox}_{\lambda f}$  iff  $0 \in \partial f(\hat{z})$   
i.e., iff  $\hat{z}$  optimal

$$\hat{z} = \text{prox}_{\lambda f}(\hat{z})$$

$$\Leftrightarrow \hat{z} \in \arg\min: f(z) + \frac{1}{2\lambda} \|z - \hat{z}\|_2^2$$

$$\Leftrightarrow 0 \in \partial f(\hat{z}) + \cancel{\frac{1}{\lambda} 0}$$

## Contractive and Non-expansive Operators

$T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  single or possibly multiple valued

$$\text{Fix } T = \{x : Tx = x\}$$

Q: When does iteration:  $x_{t+1} \leftarrow Tx_t$   
converge to a fixed point?

• If  $T$  is a contraction:

$$\textcircled{*} \|Tx - Ty\|_2 \leq L \cdot \|x - y\|_2 \text{ for } L < 1.$$

Exercise: In this case,  $Tx_t$  converges to  $\text{Fix } T$  (unique).

Exercise:  $T = (I - \lambda \nabla f)$  is a contraction when

$f$  is  $\beta$ -smooth,  $\alpha$ -strongly convex and with proper choice of  $\lambda$ .

When  $\textcircled{*}$  with  $L=1$ :  $T$  is called non-expansive. Enough? Nope.

## Averaged Operators

A non-expansive operator  $T$  is called  $\alpha$ -averaged

if:  $T = (1-\alpha)I + \alpha T'$  for some  
non-expansive operator  $T'$ .

Lemma:  $T$  is  $\alpha$ -averaged iff

$$\|Tx - Ty\|_2^2 \leq \|x - y\|_2^2 - \frac{1-\alpha}{\alpha} \|Rx - Ry\|_2^2$$

where  $R = I - T$  is the residual mapping.

Interpretation: improve by amount of fixed pt violation.

Def'n: An  $\alpha$ -averaged operator for  $\alpha = \frac{1}{2}$  is  
called firmly non-expansive.

## Convergence of Averaged Operators

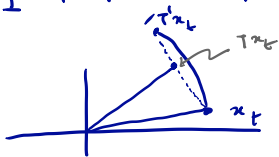
If  $T$  is  $\alpha$ -averaged,  $\alpha < 1$ ,  $\text{Fix } T \neq \emptyset$

$$x_{t+1} \leftarrow Tx_t$$

$$(a) \|x_{t+1} - x^*\| \leq \|x_t - x^*\|$$

$$(b) \|x_{t+1} - x^*\|^2 \leq \|x_0 - x^*\|^2 - \frac{1-\alpha}{\alpha} \sum_{i=1}^t \|Rx_i\|^2$$

$$T = (1-\alpha)I + T^1 \quad T^1 \text{ is non-expansive.}$$



\* Note:  $T^1 \in T$  have the same fixed points.

A consequence: If  $T$  is your fixed point operator and it's non-expansive

then:  $x_{t+1} = (1-\lambda_t)x_t + \lambda_t T x_t$ , then  $x_t \rightarrow \text{Fix } T$ .

## Compositions

- $T_i$  is non-expansive

$$\Rightarrow T = T_1 \circ T_2 \circ \dots \circ T_k \text{ is non-expansive.}$$

- $T_i$  is non-expansive, and at least one is contractive, then  $T$  is contractive,

- $T_i$  is  $\alpha_i$ -averaged, then

$$T \text{ is } \frac{k}{k-1 + \frac{1}{\max \alpha_i}} - \text{averaged}$$



## Familiar Examples

① If  $f$  is convex & smooth

$$T = (I - \lambda \nabla f) \quad \text{is averaged}$$

for right choice of  $\lambda$ .

(show).

$$\textcircled{2} \quad T = (I + \lambda \partial f)^{-1} \quad (\text{PPA})$$

$T$  is  $\alpha$ -averaged for  $\alpha = 1/2$  (i.e.,  $T$  is firmly non-expansive).

Combining ①, ② & prev.  $\S$  less:  $T = (I + \lambda \partial h)^{-1} \circ (I - \lambda \nabla g)$   
is an averaged operator  $\Rightarrow$  Prox Grad Converges.